Coherence time in biological oscillator assemblies bounds the rate of state registration

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Abstract

The coherence time of a biological oscillator assembly sets the pace at which coordinated activity becomes a registered state. This creates a speed–flexibility trade-off: independently variable phase relations expand the assembly's repertoire while increasing the coordination needed for a joint readout. We derive a bound on registration rate from the probability of satisfying the readout's requirements together, and express coherence time as a first-passage problem in relative-phase space. The resulting framework connects neural timing to target accessibility, opportunities for alignment and the survival of alignment through readout. Existing Kuramoto simulations show a strong fitted relationship in modular networks and a weak one in sparse networks, exposing the role of architecture. An illustrative neural calculation gives a 39 ms characteristic wait; narrowing the phase window raises it to 163 ms. Structured phase dynamics between registrations provide a further computational resource: they can shape which outcome becomes available at the next readout. The framework links the speed of biological processing to the organisation of its internal dynamics and yields direct tests through registration survival curves, phase perturbations and measures of temporal experience.

Keywords: biological oscillations; coherence time; coordination depth; high-dimensional dynamics; state registration; temporal perception

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1 Biological processing has a coordination timescale

Coherence time bounds how fast a biological oscillator assembly can register a new state. When a readout requires joint coordination, the assembly must reach a usable configuration and sustain it through the readout. Increasing the freedom of its phase relationships expands the available repertoire but makes a specified joint configuration harder to reach. This speed–flexibility trade-off connects the organisation of biological dynamics to the pace of neural processing.

A visual scene recruits populations that respond to different features and occupy different cortical areas. Their activity must influence downstream circuits at useful times. Oscillatory phase affects when a population is excitable and when an arriving signal is effective. This is the biological motivation for communication through coherence [1, 2, 3]. Working memory also involves temporally organised bursts and interactions across frequency bands [4, 5, 6]. A circuit's processing speed

therefore depends on the timing of relations among populations as well as on the speed of its individual components.

We call a state registration a *commit*: a transition that makes a distributed configuration available as an output with sufficient persistence to affect subsequent activity. Examples include a stable population readout, a coordinated volley that crosses a downstream threshold, or entry into a state maintained by recurrent activity.

The high-dimensional view of biological intelligence [7] places the accessible repertoire at the centre of computation. Here it also explains a timing cost: a readout requiring several relationships within restricted ranges selects a small target in that repertoire. The geometry of this target and the dynamics of reaching it determine the waiting time.

The activity between registrations remains computationally important. Structured phase relationships can retain recent inputs, shape excitability and bias the next outcome. We develop the registration bound, examine its oscillator evidence, and use it to connect continuous internal dynamics with biological timing and temporal experience.

2 From distributed phases to registered states

2.1 The configuration and the target

Let M be the number of participating modules, each an assembly with a meaningful aggregate phase. For module m, containing n_m oscillators with phases $\theta_j(t)$ at time t, define

$$z_m(t) = \frac{1}{n_m} \sum_{j \in m} e^{i\theta_j(t)} = r_m(t) e^{i\psi_m(t)}. \tag{1}$$

Here i is the imaginary unit, r_m is within-module coherence between zero and one, and ψ_m is the aggregate phase. A phase estimate becomes unreliable when r_m is too small. Module selection and the minimum acceptable coherence must therefore precede analysis of inter-module alignment.

For internally coherent modules, the inter-module order parameter is

$$z_{\text{inter}}(t) = \frac{1}{M} \sum_{m=1}^{M} e^{i\psi_m(t)} = r_{\text{inter}}(t) e^{i\bar{\psi}(t)}. \tag{2}$$

Its magnitude r_{inter} measures instantaneous phase concentration; $\bar{\psi}$ is the circular mean phase. It differs from the order parameter computed by pooling all individual oscillators.

Choose a within-module threshold r_{thr}, a full phase tolerance ε in radians, and a dwell duration D in seconds. The alignment target A_ε consists of configurations for which every $r_m \geq r_{\text{thr}}$ and

$$\max_m |\operatorname{wrap}(\psi_m - \bar{\psi})| \leq \varepsilon/2. \tag{3}$$

The function wrap maps a phase difference into $(-\pi, \pi]$. Configurations with undefined circular mean are outside the target. A qualifying episode starts at a time t for which the target condition holds throughout $[t, t + D]$; registration is completed at $t + D$. Repeated samples during the same episode count once. A new event requires departure and rearming, or a separately specified reset.

For biological registration, the target and dwell must be linked to the receiving circuit's response. Preferred nonzero phase lags and cross-frequency relationships define alternative targets. The maximum-deviation criterion checks every module; high inter-module coherence alone can conceal an outlier.

The phase configuration lives on a product of circles. Removing the common phase leaves $M-1$ relative-phase coordinates. This quotient geometry supplies $M-1$ coordinates; the dynamics determine their statistical dependence. For a smooth, positive density near complete alignment, a sufficiently small target has probability proportional to ε^{M-1} at fixed M. Confinement to a lower-dimensional support can change that exponent. Preferred offsets, target geometry and inaccessible regions can change the result further.

2.2 A rate bound from necessary coordination

To connect geometry to time, an *attempt* must represent a distinct opportunity for registration. Its definition includes which state starts the opportunity and when another opportunity becomes available. It may be a reset cycle, an independently gated sample, or a separated episode of exploration. The attempt rate counts physical opportunities independently of the recording frequency.

Suppose an attempt requires k coordination conditions. Let p_j be an upper bound on the probability of condition j being satisfied, conditional on the earlier conditions and on the relevant attempt history. The chain rule gives a bound on joint alignment, $p_A \le \prod_{j=1}^{k} p_j$, where p_A denotes alignment probability at an attempt. Let q bound the conditional probability that the alignment lasts for D and meets any remaining registration requirement. If attempts occur at asymptotic mean rate λ_a per second, then

$$\boxed{\Gamma_{\mathrm{reg}} \le \lambda_a q \prod_{j=1}^{k} p_j \le \lambda_a q\, \bar{p}^{\,k}, \qquad p_j \le \bar{p} < 1.} \tag{4}$$

Here Γ_{reg} is the long-run mean registration rate, and $\bar{p}$ is a common upper bound on conditional success. The bound assumes at most one counted registration per attempt and finite mean attempt rate. Summing the conditional expected successes over attempts establishes it even for dependent attempts. The probability bounds apply uniformly over the histories being counted.

Equation 4 expresses the core coordination constraint. Adding requirements whose conditional chances remain below one makes successful registration exponentially scarce. Coupling can make a later condition almost certain once earlier ones are met; directed control can steer the state towards the target; hierarchy can replace one large joint requirement with several smaller ones. These mechanisms reduce the coordination cost. The number of acceptable outputs also matters: registration into any of many targets can be faster than entry into one specified target. Comparisons across M hold the readout task and its allowed outcomes comparable.

For an explicit waiting-time model, assume that eligible attempts have alignment probability p_A and persistence probability $q = \mathbb{P}(\text{persistence for } D \mid \text{alignment at an attempt})$. Eligibility here includes the within-module coherence requirement; its availability is part of the attempt process.

Proposition 1 (Independent attempt model)**.** *Suppose complete attempts occupy equal intervals of duration Δ, including any required dwell and reset, and have independent success probability $s = p_A q > 0$. Starting immediately before a new attempt, the time T to the first completed success satisfies*

$$\mathbb{P}(T > n\Delta) = (1-s)^n, \qquad \mathbb{E}T = \frac{\Delta}{s}, \qquad n = 0, 1, 2, \ldots. \tag{5}$$

The number of attempts is geometric: the first n must all fail for the wait to exceed $n\Delta$. Summing this survival probability proves the mean. With $\lambda_a = 1/\Delta$, the mean is $(\lambda_a p_A q)^{-1}$. The same mean follows for independent, identically distributed complete renewal cycles with finite mean duration

Δ, even when duration and success within a cycle are associated; the survival law then need not be geometric.

For rare successes with weak dependence between opportunity clusters, a constant-hazard approximation gives, with conditional success rate h per second,

$$S(t) = \mathbb{P}(T > t) \simeq e^{-ht}, \qquad \tau_{\text{coh}} = h^{-1}, \qquad t_u = -\frac{\log(1-u)}{h}. \tag{6}$$

Here τ_{coh} is the characteristic mean waiting time and t_u is the time by which a fraction u of events has occurred. The median is $\tau_{\text{coh}} \log 2$ and the 95th percentile is approximately $3\tau_{\text{coh}}$. Short waits retain nonzero probability: the mean is not a hard minimum. The positive dwell supplies a minimum time from eligible alignment onset to completed readout. Where the model describes the start of a successful episode, that dwell must be added to its timestamp.

2.3 Dependence, entry and persistence

Coherence time is a target-specific waiting time. Stationary occupancy combines entry frequency and residence duration: rare, long visits and frequent, short visits can produce the same occupancy. Full-process mixing also measures something different. A slowly relaxing variable unrelated to the target can lengthen mixing while leaving entry unchanged. Recurrence identities for stationary discrete processes [8] apply to their specified recurrence experiment; continuous-time first entrance requires its own initial state and entry dynamics.

Measure λ_a for the target and opportunity definition in use. An inverse autocorrelation time of r_{inter} provides a proxy after calibration against actual opportunities, including their clustering. Its scalar summary discards phase directions. Angular frequency spread and carrier frequency supply further timescales; neither is generally equal to the target's attempt rate.

One empirical alternative is to count separated entries into the target and measure the fraction that survive the dwell. If ν_A is the long-run entry rate and q_{ent} the entry-conditioned survival fraction, then $\nu_A q_{\text{ent}}$ is the rate of qualifying episodes, subject to the rearming rule. The first-passage distribution additionally depends on preparation: arrival from an experimental reset, recurrence after a commit and forward waiting from a randomly chosen time have different initial conditions and need not be exponential.

Persistence can contribute as much to scaling as entry. For example, if each of M aligned modules independently exits at rate β, then

$$q(M, D) = e^{-M\beta D}. \tag{7}$$

The exit rate β has units of inverse seconds. In this example $-\log q = M\beta D$, so dwell adds a term growing linearly with M to the logarithm of the mean wait. Use a constant persistence factor over a range where measured survival remains approximately constant.

For a deterministic covariate schedule, or an exogenous one conditional on its path, the survival expression is $S(t \mid \mathcal{H}) = \exp[-\int_0^t h(v \mid \mathcal{H})\, dv]$, where $\mathcal{H}$ denotes that history and v is elapsed time. Pooling heterogeneous histories can broaden the waiting distribution. Binning events by coherence measured at success selects unusually aligned states; estimate pre-event hazards from the full risk set. Endogenous phase histories require a joint model of state and event dynamics. Correlated attempts, metastability and slow modulation should be compared using the full survival data, including trials with no observed registration.

3 Alignment probability and high-dimensional flexibility

3.1 A tractable phase-distribution model

For a single relative-phase coordinate ϕ, a useful closure is the von Mises density

$$f_\kappa(\phi) = \frac{e^{\kappa\cos\phi}}{2\pi I_0(\kappa)}, \qquad p_1(\varepsilon,\kappa) = \int_{-\varepsilon/2}^{\varepsilon/2} f_\kappa(\phi)\,d\phi. \tag{8}$$

The concentration $\kappa \geq 0$ controls the spread around zero; I_0 and I_1 are modified Bessel functions of the first kind. For this distribution, its population mean resultant length r satisfies $r = I_1(\kappa)/I_0(\kappa)$ [9]. The von Mises density maximises entropy given the first circular moment. For the closure, use the resultant length of the specified relative-phase distribution. Instantaneous r_{inter} and its trajectory average are different statistics. Substituting trajectory-average coherence for r requires a distributional check using all relevant trajectories.

For $M-1$ independent anchored relative phases, each with density f_κ, the rectangular alignment target has exact probability $p_A = p_1^{M-1}$. The mean-centred target in Eq. 3 has a different shape and dependent deviations. Its finite-window probability needs a geometry factor that can vary with M, even where the local exponent is $M-1$.

For a declared range of tolerances and network conditions, consider the empirical approximation

$$p_A \simeq C_M p_1^d, \qquad d = \alpha(M-1), \qquad \tau_{\text{coh}} \simeq \frac{p_1^{-\alpha(M-1)}}{\lambda_a C_M q}. \tag{9}$$

The factor C_M accounts for target geometry and residual distributional effects; d is a fitted probability exponent and α its value per relative-phase coordinate. Require the resulting probability to lie between zero and one. The range $0 < \alpha \leq 1$ describes dependencies that facilitate joint success relative to the product reference; general dependencies allow other exponents. Estimate C_M across module counts. The empirical closure and the history-uniform bounds in Eq. 4 have separate evidential requirements.

At fixed κ and small ε, $p_1 \simeq \varepsilon e^\kappa/(2\pi I_0(\kappa))$. Thus a constant d gives an inverse-power dependence on tolerance. Across M, the logarithmic form is

$$\log\tau_{\text{coh}} \simeq -\log(\lambda_a C_M q) + \alpha(M-1)[-\log p_1]. \tag{10}$$

An exponential dependence on module count is predicted when α and p_1 stay approximately fixed and the remaining factors do not compensate exponentially.

3.2 Dimension depends on the question and the scale

The directly relevant local dimension of the target probability is the tolerance slope

$$d_A(\varepsilon) = \frac{d\log p_A(\varepsilon)}{d\log\varepsilon}. \tag{11}$$

This quantity measures how rapidly acceptable configurations become scarcer as the required precision increases, holding the preparation and attempt rule fixed. At finite tolerance it can vary with ε. It agrees with d in Eq. 9 only where the closure, its constant factors and its small-window approximation are adequate. The geometry of the support controls the asymptotic exponent when a regular local density exists.

Covariance gives another useful view. For a specified real coordinate representation of the relative phases, let C be its covariance and λ_ℓ its nonnegative eigenvalues. The participation ratio is

$$D_{\mathrm{PR}} = \frac{(\operatorname{tr} C)^2}{\operatorname{tr}(C^2)} = \frac{(\sum_\ell \lambda_\ell)^2}{\sum_\ell \lambda_\ell^2}. \tag{12}$$

It measures how widely variance is distributed among modes and lies between one and the covariance rank when variance is nonzero. This summarises the distribution of neural activity across modes [10, 11].

For a nonsingular Gaussian in k dimensions, a sufficiently small ball centred at a point of positive density has probability proportional to its radius to the power k, however unequal the covariance eigenvalues. Anisotropy changes its coefficient and the scales over which directions become resolvable. Participation ratio consequently neither supplies nor generally bounds the target exponent. It can be tested as an empirical predictor of d_A over a stated tolerance range. Noise can increase participation ratio as well as structured activity.

For circular data, report the representation and scale: $M - 1$ anchored phase differences, a local unwrapped chart, or a sine–cosine embedding. Wrapped deviations from the circular mean do not generally sum to zero as real numbers. Determine covariance rank from the chosen coordinates.

3.3 The speed–flexibility trade-off

The model gives a physical reason for local fast processing and slower distributed integration to coexist. If each additional independent phase relation can occupy several distinguishable ranges, adding relations expands the available configuration repertoire. Requiring one particular combination makes joint success less likely. The same freedom that permits more possible combinations can therefore delay a particular registration.

Within the von Mises closure, increasing concentration raises p_1 and speeds alignment at fixed attempt rate, dwell and target. Tight phase locking also suppresses variation in relative phases. At finite noise, concentration can reduce the accessible repertoire at the readout's resolution without changing the mathematical rank of the support. Amplitude variation, within-module dynamics and cross-frequency relations can remain rich. High coherence is thus compatible with substantial dynamical complexity outside the particular phase coordinates constrained by the readout.

Architecture controls this trade-off. Coupling changes the attempt process, persistence and target accessibility as well as concentration. Hierarchical organisation preserves local degrees of freedom while reducing the number that must coordinate simultaneously. It therefore provides a route to flexible processing with fast readout.

4 Oscillator evidence for the coordination cost

4.1 Networks, detection and fitted scaling

The existing numerical study uses noisy Kuramoto oscillators [12, 13]. With ω_i the natural angular frequency of oscillator i, K_{ij} its coupling weights and σ the noise amplitude, the dynamics have the form

$$d\theta_i = \left[\omega_i + \sum_j K_{ij} \sin(\theta_j - \theta_i)\right] dt + \sigma\, dW_i. \tag{13}$$

The W_i are independent standard Wiener processes. Coupling normalisation is included in K_{ij}. The saved comparison uses $N = 100$ oscillators, module counts $M = 3, \ldots, 10$, 20 trials per count, a

Table 1: Existing topology comparison. n is the number of module-count points with usable medians in the regression; each count has 20 trials. R^2 describes the fitted logarithmic response, not held-out survival prediction. The all-to-all exponent comes from a different, linearised predictor and is not directly comparable to the other two.

Topology	Predictor	$\hat{\alpha}$	R^2	n
Modular	Exact p_1 closure	0.989	0.975	5
All-to-all	Linearised surrogate	1.137	0.779	8
Sparse random	Exact p_1 closure	0.121	0.279	4

60 s observation horizon and a 5 ms integration step. The natural-frequency standard deviation is 0.3 rad s^{-1} and the numerical noise amplitude is 0.1 rad $s^{-1/2}$. All three constructions use overall coupling 0.7 s^{-1} after row normalisation. Before normalisation, modular within-module edges have weight 1.0 and between-module edges weight 0.08. The sparse connection probability is 0.03.

The detector requires every within-module coherence to exceed 0.6 and every module phase to lie within 0.5 radians of the circular mean, giving full tolerance $\varepsilon = 1.0$ rad. It requires six consecutive qualifying samples, implementing a nominal 30 ms dwell at the 5 ms step. The saved event time is the start of that run. These are therefore waiting times to successful alignment onsets from random initial phases. Completed readout includes the dwell; stationary intervals require repeated events. The detector verifies persistence at the sampled integration points.

Initial phases and natural frequencies are drawn afresh across trials; sparse adjacency matrices are also redrawn. Event detection starts immediately; trajectory-average coherence excludes the first quarter of each trajectory. Non-hits are right-censored at 60 s. The fits use finite Kaplan–Meier medians, so not every module count contributes. Table 1 reports the saved fit summaries.

For the exact-p_1 regressions, the predictor is $(M-1)[-\log p_1(\varepsilon, \kappa(\bar{r}_{\text{inter}}))]$, where $\bar{r}_{\text{inter}}$ is trajectory-average coherence pooled over all trials. The response is the logarithm of the product of the median waiting time and the autocorrelation-based attempt proxy. An intercept is fitted. For all-to-all coupling, the mean window probability is about 0.998, leaving little variation in the exact predictor; the reported surrogate instead uses $(1-\bar{r}_{\text{inter}})(M-1)\log(2\pi/\varepsilon)$.

The modular network shows a strong fitted coordination cost ($R^2 = 0.975$). The sparse network's weak fit ($R^2 = 0.279$) exposes the importance of architecture. All-to-all coupling explores a near-synchronised regime in which alignment is readily accessible. Together, these results place partially coordinated modular dynamics at the centre of the present empirical case.

The saved fits use an attempt proxy averaged over hit trials and finite medians from the less censored conditions. These selections can affect the slope. The free intercept combines the mean–median conversion, target geometry and persistence. Independent estimates of those factors would turn the fitted relationship into a survival prediction.

At fixed total N, increasing M also reduces module size, bringing finite-size fluctuations and effective coupling into the comparison. The accepted study reports within-module mean coherence between 0.66 and 0.74 without a declining trend, and less than 5% change in median waits on halving or doubling the nominal dwell. These results demonstrate robustness over the tested range; varying module count at fixed module size would isolate the effect of coordination depth.

The existing exponentiality diagnostic groups observed hits by trajectory-average coherence. Its small groups describe the observed subset. The next test uses the full censored survival curves and predicts outcomes from a stated initialisation.

Table 2: Sensitivity of the illustrative closure. All rows use $\alpha = 0.7$ and $C_M q = 1$. Probabilities use numerical inversion and quadrature, without a small-window approximation. The entries are characteristic waiting times; any additional readout delay is separate.

Scenario	M	r	ε	λ_a (s^{-1})	τ_{coh} (ms)
Base	8	0.75	$\pi/2$	126	39
Fewer modules	6	0.75	$\pi/2$	126	25
More modules	10	0.75	$\pi/2$	126	61
Lower concentration	8	0.65	$\pi/2$	126	75
Higher concentration	8	0.85	$\pi/2$	126	19
Narrower window	8	0.75	$\pi/3$	126	163
Faster attempts	8	0.75	$\pi/2$	189	26

4.2 An illustrative neural timescale

Neural timing studies place perceptual integration on a scale of tens of milliseconds [14, 15, 16]. The phase model reaches this scale with a compact illustrative parameter set. Behavioural thresholds additionally include stimulus processing, decision requirements and readout persistence.

Consider the reduced closure with $C_M q = 1$, $M = 8$, $\alpha = 0.7$, $r = 0.75$, full tolerance $\varepsilon = \pi/2$ and attempt proxy $\lambda_a = 126\ s^{-1}$. These values define a scenario rather than an independently measured human assembly. Inverting the Bessel-function ratio and integrating Eq. 8 gives

$$\kappa = 2.369301, \qquad p_1 = 0.723078, \qquad \tau_{coh} = \frac{p_1^{-4.9}}{126} = 0.038871 \text{ s}. \tag{14}$$

The product approximation produces a characteristic wait of 39 ms. Its alignment probability is about 0.204, outside a deep rare-event regime, so an exponential survival law needs testing. This neural scenario is separate from the seconds-scale numerical network sweep.

Narrowing the window to $\pi/3$ gives $p_1 = 0.539752$ and $\tau_{coh} = 162.883$ ms. Joint variation is broader still: at fixed $\alpha = 0.7$ and $\varepsilon = \pi/2$, the combinations $(M, r, \lambda_a) = (6, 0.85, 189)$ and $(10, 0.65, 63)$ give 9.768 and 282.686 ms, respectively. These scenario endpoints show how architecture and coordination shift the timescale; confidence limits require a joint uncertainty analysis. Measuring the attempt process, target geometry and dwell survival converts this calculation into a prediction for a particular assembly.

5 Predicting registration from architecture

The central experimental prediction links architecture, precision and timing. Estimate the coordination model on one dataset and predict the survival curve on another. A single behavioural waiting time leaves λ_a, C_M, q and α underdetermined.

Specify the biological output and detector: module membership, frequency bands, within-module threshold, phase tolerance, dwell and rearming. Calibrate these against the receiving circuit's response in separate data. A cross-correlogram half-width can inform a tolerance: temporal half-width w at angular frequency ω_0 corresponds to full phase width $2\omega_0 w$. The receiving circuit determines which width matters. Control volume conduction and shared inputs when measuring coordination.

Estimate alignment probabilities across tolerances using eligible opportunities from all trials. Measure target-entry rate, persistence and temporal clustering separately, and test the attempt

proxy across conditions. Measure participation ratio and target exponent independently; train any mapping between them before predicting the behavioural window.

Predict the held-out survival curve, including non-hits. Retain censored trials in the likelihood and compare constant-hazard, renewal and state-dependent-hazard models under the experimental initial state. Propagate joint uncertainty in module selection, distribution parameters, attempt rate and persistence, preserving their covariance.

Varying M at approximately fixed module size and within-module organisation isolates the coordination cost. Varying tolerance at fixed opportunity process isolates the precision cost. Both manipulations predict slower registration as the joint requirement becomes harder to satisfy. Successful survival prediction across architectures would establish the explanatory reach of the framework; systematic failure after independent parameter measurement would count against it.

6 Physical limits and the cost of a readable outcome

Coordination places a macroscopic timing constraint on assemblies whose individual components operate much faster. An event's latency must also exceed every necessary physical stage: sequential stages add, while concurrent stages overlap.

For an isolated quantum system evolving to an orthogonal state under a time-independent Hamiltonian, the Mandelstam–Tamm and Margolus–Levitin bounds take the form

$$\tau_{\perp} \geq \max\left(\frac{\pi\hbar}{2\Delta E}, \frac{\pi\hbar}{2(\langle E\rangle - E_0)}\right). \tag{15}$$

Here $\hbar$ is the reduced Planck constant, ΔE the energy standard deviation, $\langle E\rangle$ the mean energy and E_0 the ground-state energy [17, 18]. Choosing both energy scales near $k_{\mathrm{B}}T_b$, with Boltzmann constant k_{B} and temperature $T_b = 310$ K, gives roughly 4×10^{-14} s. This illustrative microscopic floor lies far below millisecond coordination. A neural assembly's open-system speed limit instead depends on its energy distribution and interaction with its environment.

Noise sets a further integration timescale. Detecting a weak signal takes time even when populations already align; the signal model, noise spectrum and decision criterion determine that delay.

If each completed operation dissipates mean energy E_{reg} and available sustained power is P, then $\Gamma_{\mathrm{reg}}E_{\mathrm{reg}} \leq P$. For a memory reset to a standard state, erasing b unbiased independent bits in the usual isothermal setting dissipates at least $bk_{\mathrm{B}}T_b \ln 2$ [19]. The memory states, their probabilities and the reset operation specify this cost. Participation-ratio changes and phase threshold crossings do not identify erased bits.

A low-dimensional readout can leave the underlying high-dimensional state active or transfer information elsewhere. Measure the physical readout and reset costs alongside coordination and noise to identify the assembly's rate-limiting step.

7 Computation between registrations

7.1 Structured phase offsets

The interval before registration is a computational resource. Coupled assemblies can maintain preferred relative phases that encode recent inputs, shape excitability and alter which target is reached next. We propose *phase-delta dynamics* as the mechanism: structured relative-phase evolution outside the registration target that influences future outputs.

For modules m and n, their phase offset is $\Delta\psi_{mn}(t) = \mathrm{wrap}(\psi_m(t) - \psi_n(t))$. A computational phase pattern carries information about future outcomes beyond shared inputs, current amplitude and other measured state variables, and causally influences those outcomes. Persistently locked nonzero offsets can produce low global coherence with little exploration; the evolving offset structure supplies the relevant evidence.

Different configurations can share the same global coherence yet favour different outcomes. This allows a concrete test: perturb the offset structure while matching the measured alignment opportunity rate, dwell survival and relevant amplitude statistics as closely as possible. A change in outcome accuracy or identity beyond that explained by altered registration timing would support a computational role for the pre-registration configuration. Holding mean coherence fixed alone is insufficient, because the target probability can still change.

This view extends communication through coherence [1, 2]. The relation to a receiver governs when a population communicates; the path through phase space also shapes what becomes available at readout. Continuous computation and communication along that path prepare the registered outcome.

7.2 Physical substrates and spatial extent

Phase describes physical activity sustained by synaptic interaction, subthreshold membrane dynamics and dendritic integration. Extracellular fields offer a further route to continuous coordination: ephaptic effects influence cortical neurons and cerebellar spike timing [20, 21]. Field-mediated organisation has also been proposed at broader scales [22]. Comparing field, synaptic and common-input contributions connects the phase description to its biological machinery.

A spatially extended narrowband signal can project many interacting modes onto a simple waveform. A large population moving in lockstep, by contrast, has few independent relative-phase directions. Measuring spatial coverage, covariance structure and target probability together distinguishes these organisations.

Under a constant-hazard model, the probability of at least one registration within a cycle of frequency f is $1 - e^{-h/f}$. To achieve reliability u per cycle requires $h \geq -f \log(1 - u)$. At fixed attempt rate and with a readout required every cycle, faster rhythms demand a more accessible target or more persistent alignment. This timing constraint ties frequency to the coordination task; frequency alone supplies no universal ordering of alpha and gamma dimensionality.

7.3 Temporal experience and dynamics between registrations

The framework separates the pace of registered outcomes from the richness of the dynamics between them. This distinction offers a physical route to understanding why temporal resolution, experienced duration and retrospective duration judgements differ.

Alpha frequency has been associated with two-flash discrimination and crossmodal temporal integration [14, 15]. For a rhythm that gates opportunities with λ_a proportional to frequency and fixed $p_A q$, the prediction is $\tau_{\mathrm{coh}} \propto 1/f$. A 20% frequency increase then reduces the characteristic wait by about 16.7%. Entrainment studies [23] provide an experimental route: measure concentration, persistence and decision criteria alongside frequency to distinguish their contributions.

We propose that experienced time depends partly on structured dynamics between registrations. Richer exploration makes more relationships available within each interval, even at an unchanged registration rate. This predicts that structured dynamical richness contributes to prospective duration judgements after accounting for registration timing, attention and memory formation.

Stress exposes the distinction between duration and resolution. In a free-fall experiment, participants retrospectively judged a frightening event as longer without improved temporal resolution [24]. Denser memory encoding explains this dissociation. A coordination account predicts an additional contribution only when stress changes the relevant architecture or waiting distribution; measuring those changes separates the mechanisms across perceptual and motor pathways.

Pharmacological perturbations provide another test. Increased signal diversity under psychedelic drugs [25, 26] and the entropic-brain and REBUS accounts [27, 28] connect altered neural organisation with altered experience. The present proposal predicts that time dilation tracks reproducible, structured variation beyond any change in registration rate. Compare prospective duration judgements with retrospective reports while measuring both variables. This distinguishes computational richness from unstructured noise, which also raises signal diversity and participation ratio.

8 Biological implications

Coherence time links the speed of biological processing to the freedom of its internal dynamics. For functional events requiring simultaneous conditions in an identifiable state space, the same coordination argument extends beyond neural oscillators. Spatial, mechanical and biochemical constraints define their own targets and entry processes. This provides a common way to investigate timing in distributed biological assemblies.

Architecture offers a way to manage the cost. Local handshakes, directed coordination and hierarchical readouts reduce the number of requirements that must be satisfied together. These principles apply to engineered systems as well: an asynchronous design pays for rare joint requirements, while a global clock supplies a shared timing structure. The organisation of the readout determines how much coordination each output demands.

Scope and open questions. The main empirical task is to predict registration survival across architectures from independently measured entry and persistence dynamics. The phase-delta and temporal-experience proposals add causal questions: which offset structures change the registered outcome, and which aspects of the intervening dynamics change experienced duration? The perturbation and timing experiments above make these questions testable.

The speed–flexibility trade-off gives coherence time its biological significance. Independent phase relationships expand the repertoire; joint readout places a timing cost on accessing that repertoire. Coupling, targeting and hierarchy control the balance. Biological assemblies can therefore sustain rich internal dynamics while their coordination architecture sets the pace at which those dynamics become usable new states.

Data and code

The accompanying accepted source and existing simulation code are retained alongside this edition. The calculations in Table 2 use Eq. 8 and Eq. 9. The topology summaries use the saved validation outputs; no new network simulations are reported. The project repository is `https://github.com/todd866/coherence-time-biosystems`.

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